Keru Chen

PhD Student, Electrical Engineering, Arizona State University
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Research focus: Reinforcement learning, LLM post-training, and AI agents, including multi-agent systems; safety, alignment, and constrained decision-making, with interest in healthcare applications.

EDUCATION

Arizona State University Aug 2025 - Present
PhD in Electrical Engineering
Advisor: Prof. Shaofeng Zou

Xi'an Jiaotong University 2021 - 2025
BEng in Automation

PUBLICATIONS & PREPRINTS

Information-Directed Offline-to-Online Reinforcement Learning
Keru Chen
NeurIPS 2026, Main Conference. Also at ICML 2026 DEMO Workshop.

HIPO: Instruction Hierarchy via Constrained Reinforcement Learning
Keru Chen, Jun Luo, Sen Lin, Yingbin Liang, Alvaro Velasquez, Nathaniel Bastian, Shaofeng Zou
EMNLP 2026, Main Conference.

Towards Fast Safe Online Reinforcement Learning via Policy Finetuning
Keru Chen, Honghao Wei, Zhigang Deng, Sen Lin
Transactions on Machine Learning Research (TMLR), 2026.

Fourier Ptychographic Microscopy 10 Years On: A Review
F. Xu, Z. Wu, C. Tan, Y. Liao, Z. Wang, **Keru Chen**, An Pan
Cells, 13(4), 324, 2024.

Preprints

Training LLMs for Multi-Agent Collaboration with Verifiable Information Rewards
Keru Chen, Sen Lin, Yingbin Liang, Nathaniel D. Bastian, Shaofeng Zou
arXiv:2610.17893, 2026.

MeshHeal: Two-Timescale Self-Healing for Gray Failures in Decentralized LLM Agent Networks
Keru Chen, Sen Lin, Yingbin Liang, Nathaniel D. Bastian, Shaofeng Zou
arXiv:2609.29015, 2026.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB. **Deep learning:** PyTorch.
LLM Frameworks: Transformers, vLLM, verl.
Tools: Git/GitHub, Linux, LaTeX.

RESEARCH EXPERIENCE

Arizona State University Aug 2025 - Present

Doctoral Research | Prof. Shaofeng Zou

- Post-trained LLMs for collaboration using surprisal-reduction rewards in a verifiable toy environment, improving performance on out-of-distribution collaboration benchmarks (**CoSIR; preprint**).
- Formulated instruction hierarchy in LLMs as constrained RL (**HIPO; EMNLP 2026**).
- Developed information-directed exploration for offline-to-online RL (**NeurIPS 2026**).
- Studied peer review and two-timescale monitoring to detect unreliable outputs and guide recovery in decentralized LLM agent networks (**MeshHeal; preprint**).
- Work closely with Prof. Sen Lin and Prof. Yingbin Liang.

University of Houston Sep 2023 - Jul 2025

Research Intern | Prof. Sen Lin

- Studied value alignment and adaptive constraint penalties for safe online fine-tuning of offline RL policies (**first-author paper, TMLR 2026**).

Westlake University Summer 2024

Research Intern | Prof. Donglin Wang

- Applied reinforcement learning to fine-tune vision-language-action models for robotic manipulation.

University of North Carolina at Chapel Hill Jan 2024 - May 2024

Research Intern | Prof. Tianlong Chen

- Worked on time-series forecasting and security analysis of LLM and retrieval-augmented generation systems; contributed to research published at EMNLP 2024.

RESEARCH SUPPORT

OpenAI Researcher Access Program Oct 2026

ACADEMIC SERVICE

Conference Reviewer: NeurIPS 2026; ICLR 2027.
Journal Reviewer: Scientific Reports; IEEE/ACM Transactions on Networking.